

The Normal Distribution

Anaya Scott

Question 1

0/1 pt 3 19

For a standard normal distribution, find:

$$P(z > c) = 0.7297$$

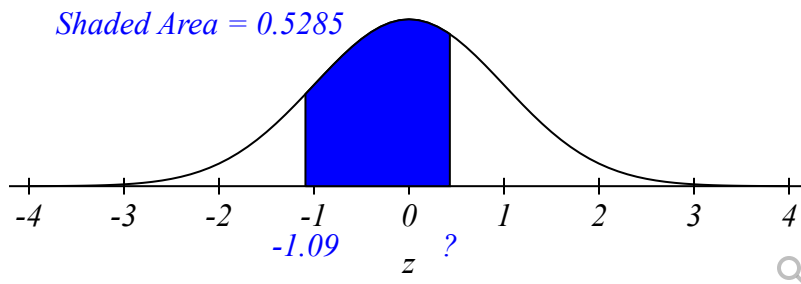
Find c rounded to two decimal places.

Question Help: [Video 1](#) [Video 2](#)

Question 2

0/1 pt 3 19

Use the given shaded area (0.5285) in the middle of the standard normal distribution and the given z-score (-1.09) to find the missing z-score. The shaded region is not symmetric about $z = 0$. Round your final answer to two decimal places.



Upper z-score = _____

Question Help: [Video](#)

Question 3

0/1 pt 3 19

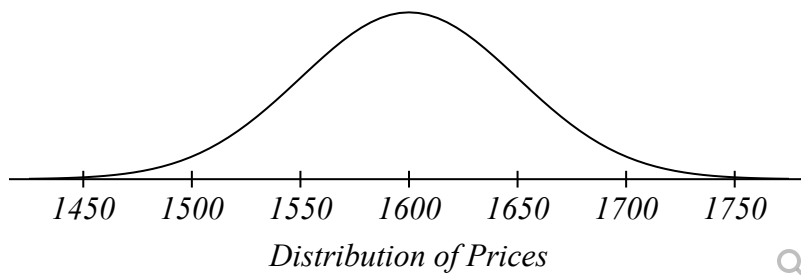
The scores on a standardized test are normally distributed with a mean of 90 and standard deviation of 10. What test score is 0.3 standard deviations above the mean?

Question Help: [Video](#)

Question 4

0/1 pt 3 19

The graph illustrates a normal distribution for the prices paid for a particular model of HD television. The mean price paid is \$1600 and the standard deviation is \$50.



What is the approximate percentage of buyers who paid between \$1600 and \$1650?

_____ %

What is the approximate percentage of buyers who paid between \$1600 and \$1750?

_____ %

What is the approximate percentage of buyers who paid between \$1550 and \$1650?

_____ %

What is the approximate percentage of buyers who paid less than \$1450?

_____ %

What is the approximate percentage of buyers who paid less than \$1500?

_____ %

What is the approximate percentage of buyers who paid between \$1500 and \$1600?

_____ %

Question Help: [Video 1](#) [Video 2](#)

Question 5

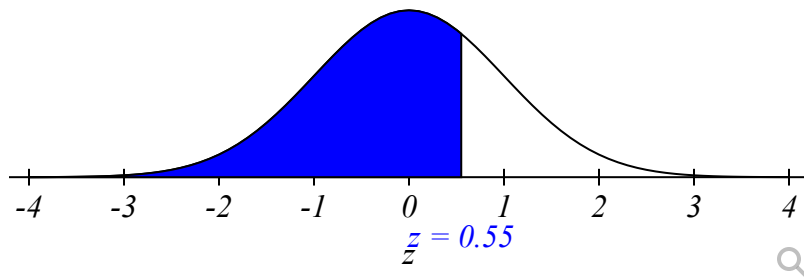
0/1 pt 3 19

The scores on the Accuplacer test and High School GPAs are normally distributed. The Accuplacer test had a mean of 40 and a standard deviation of 8. High School GPAs had a mean of 2.6 and a standard deviation of 0.3. What high school GPA do you need to equal a score of 28 on the Accuplacer test? Give answer to two decimal places.

Question 6

0/1 pt 3 19

Find the area of the shaded region under the standard normal distribution to the left of the given z-score. Round your answer to four decimal places.



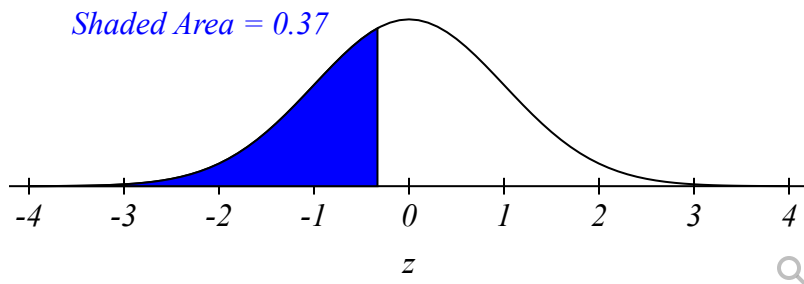
$$P(z < 0.55) = \underline{\hspace{2cm}}$$

Question Help: [Video](#)

Question 7

0/1 pt 3 19

Find the z-score for the given shaded region under the standard normal distribution. Round your answer to two decimal places.



$$\text{z-score} = \underline{\hspace{2cm}}$$

Hint: Try using [StatKey](#) or a spreadsheet, =NORMINV(p, mean, std)

Question Help: [Video 1](#) [Video 2](#)

Question 8

0/1 pt 3 19

For the standard normal distribution ($\mu = 0$ and $\sigma = 1$), find the probability that a z-score is less than -1.69:

$$P(z < -1.69) = \underline{\hspace{2cm}}$$

Hint: Try using [StatKey](#) or a spreadsheet, =NORMDIST(z-score,0,1,true).

Question Help: [Video](#)

● Question 9

0/1 pt 3 19

A normal distribution has a mean of 99 and a standard deviation of 7. Find the z-score for a data value of 92.

Use the formula $Z = \frac{x - \mu}{\sigma}$ where μ is the mean, σ is the standard deviation, and x is the data value.

_____ Round to 4 decimal places

Question Help: [Video](#)

● Question 10

0/1 pt 3 19

For a standard normal distribution, find:

$P(z < -2 \text{ or } z > -1.85)$

Question Help: [Video 1](#) [Video 2](#)

● Question 11

0/1 pt 3 19

For the standard normal variable Z , find $z_{0.3}$, the value that divides the area under the density curve into 0.7 to the left and 0.3 to the right.

$z_{0.3} =$ _____ (Round the answer to 2 decimal places)

Question Help: [Video](#)

● Question 12

0/1 pt 3 19

Find the 86th percentile for the standard normal random variable Z .

The 86th percentile for Z is _____ (Round the answer to 2 decimal places)

Question Help: [Video](#)

● Question 13

0/1 pt 3 19

Find the value that divides the area under the standard normal probability density curve into 0.67 to the left and 0.33 to the right.

$z_{0.33} =$ _____ (Round the answer to 2 decimal places)

Question Help: [Video](#)

● Question 14

0/1 pt 3 19

A manufacturer knows that their items have a normally distributed length, with a mean of 18.4 inches, and standard deviation of 2.2 inches.

If one item is chosen at random, what is the probability that it is less than 18.3 inches long?

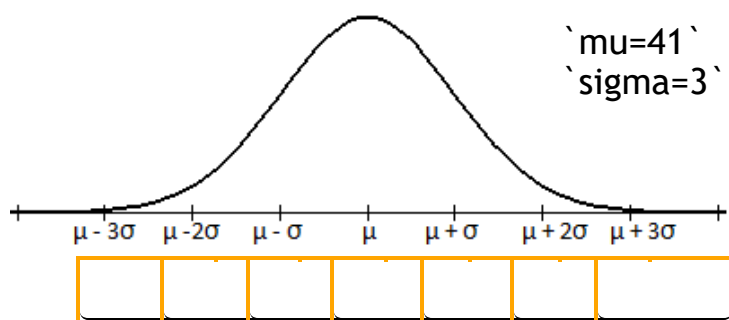
Question Help: [Video](#)

● Question 15

0/1 pt 3 19

The time to complete an exam is approximately Normal with a mean of 41 minutes and a standard deviation of 3 minutes.

The bell curve below represents the distribution for testing times. The scale on the horizontal axis is equal to the standard deviation. Fill in the indicated boxes.



Question Help: [Video](#)

● Question 16

0/1 pt 3 19

For a normal variable $X \sim N(\mu=36.1, \sigma=2.5)$, find the probability $P(X < 30.85)$:

$P(X < 30.85) =$ _____ (Round the answer to 4 decimal places)

Question Help: [▶ Video 1](#) [▶ Video 2](#) [▶ Video 3](#)

● Question 17

0/1 pt 3 19

For a normal variable $X \sim N(\mu=72.1, \sigma=11.1)$, find the probability $P(56.25 < X < 82.25)$:

$P(56.25 < X < 82.25) =$ _____ (Round the answer to 4 decimal places)

Question Help: [▶ Video 1](#) [▶ Video 2](#) [▶ Video 3](#)

● Question 18

0/1 pt 3 19

For a normal variable $X \sim N(\mu=20.9, \sigma=4.78)$, find $x_{0.52}$, the value that divides the area under the density curve into 0.48 to the left and 0.52 to the right.

$x_{0.52} =$ _____ (Round the answer to 2 decimal places)

Question Help: [▶ Video 1](#) [▶ Video 2](#) [▶ Video 3](#)

● Question 19

0/1 pt 3 19

Find the 75th percentile for a normal random variable $X \sim N(\mu=90.1, \sigma=19.9)$.

The 75th percentile for X is _____ (Round the answer to 2 decimal places)

Question Help: [▶ Video 1](#) [▶ Video 2](#) [▶ Video 3](#)

● Question 20

0/1 pt 3 19

Based on a study, it was observed that amount of blood loss during major surgery is normally distributed with a mean 286.2 ml and a standard deviation of 68.3 ml. Let X be the amount of blood loss during major surgery of a randomly selected patient.

1. Describe the probability distribution of X and state its parameters μ and σ :

 $X \sim$ ($\mu =$ _____, $\sigma =$ _____)

2. Find the probability that the amount of blood loss during major surgery of a randomly selected patient is

- a. less than 374.7 ml. _____ (Round the answer to 4 decimal places)
- b. between 426.9 and 466.3 ml. _____ (Round the answer to 4 decimal places)
- c. more than 392.2 ml. _____ (Round the answer to 4 decimal places)

3. Find the 55-th percentile for the amount of blood loss during major surgery.
_____ ml (Round the answer to 1 decimal place)

Question Help:  [Video 1](#)  [Video 2](#)  [Video 3](#)

● Question 21

☒ 0/1 pt  3  19

In New Jersey, 51 % of the population favors the incumbent for governor. A simple random sample of 450 is taken. Use the normal approximation along with the continuity correction factor to find the probability that at least 223 respondents favor the incumbent for governor.

- Let X be the number of respondents out of 450 that favor the incumbent for governor. Describe the distribution of X and its parameters:

$X \sim$ ($n =$ _____, $p =$ _____)

- Use the random variable notation to symbolically express the probability that at least 223 respondents favor the incumbent for governor:

- Let Y be a normal variable that will be used to approximate the probability in question. Find the parameters of Y (round the answers to 2 decimal places):

$Y \sim$ ($\mu =$ _____, $\sigma =$ _____)

- Use the random variable notation to symbolically express the approximate probability that at least 223 respondents favor the incumbent for governor:

- Use the correction for continuity:

- Find the probability (round the answer to 4 decimal places):

Question Help:  [Video](#)

● Question 22

 0/1 pt  3  19

Assume that IQ scores are normally distributed with a mean 101.9 pts and a standard deviation of 11.8 pts. Let X be the IQ score of a randomly selected person.

1. Describe the probability distribution of X and state its parameters μ and σ :

 $X \sim$ ($\mu =$ _____, $\sigma =$ _____)

2. Find the probability that the IQ score of a randomly selected person is

- a. less than 113.6 pts. _____ (Round the answer to 4 decimal places)
- b. between 121.1 and 125 pts. _____ (Round the answer to 4 decimal places)
- c. more than 129.6 pts. _____ (Round the answer to 4 decimal places)

3. Find the 65-th percentile for the IQ score. _____ pts (Round the answer to 1 decimal place)

Question Help:  [Video 1](#)  [Video 2](#)  [Video 3](#)

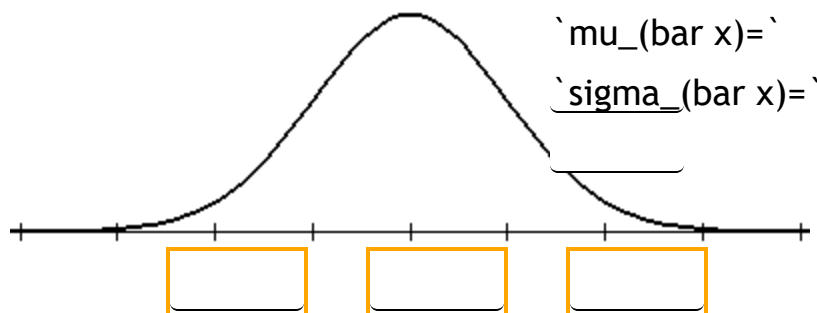
● Question 23

 0/1 pt  3  19



Let X represent the full length of a certain species of newt. Assume that X has a normal probability distribution with mean 176.4 inches and standard deviation 6.9 inches.

You intend to measure a random sample of $n=57$ newts. The bell curve below represents the distribution of these sample means. The scale on the horizontal axis is the standard error of the sampling distribution. Complete the indicated boxes, correct to **two decimal places**.



Question Help: [Video](#)

● Question 24

0/1 pt 3 19

Historically, the SAT score of a randomly selected student has unknown distribution with a mean of 1483 points and a standard deviation of 340.3 points. Let X be the SAT score of a randomly selected student and let $\bar{X}$ be the average SAT score of a random sample of size 36.

1. Describe the probability distribution of X and state its parameters μ and σ :

$X \sim$ ($\mu =$ _____, $\sigma =$ _____)

and find the probability that the SAT score of a randomly selected student is between 897 and 1458 points.

_____ (Round the answer to 4 decimal places)

2. Use the Central Limit Theorem

to describe the probability distribution of $\bar{X}$ and state its parameters $\mu_{\bar{X}}$ and $\sigma_{\bar{X}}$: (Round the answers to 1 decimal place)

$\bar{X} \sim$ ($\mu_{\bar{X}} =$ _____, $\sigma_{\bar{X}} =$ _____)

and find the probability that the average SAT score of a sample of 36 randomly selected students is less than 1339 points.

_____ (Round the answer to 4 decimal places)

Question Help: [Video](#)

● Question 25

0/1 pt 3 19

According to a study, the weight of a randomly selected adult female is normally distributed with a mean of 178 lbs and a standard deviation of 8.8 lbs. Let X be the weight of a randomly selected adult female and let S be the total weight of a random sample of size 28.

1. Describe the probability distribution of X and state its parameters μ and σ :

$X \sim$ ($\mu =$, $\sigma =$)

and find the probability that the weight of a randomly selected adult female is more than 198 lbs.

(Round the answer to 4 decimal places)

2. Use the Central Limit Theorem

to describe the probability distribution of S and state its parameters μ_S and σ_S : (Round the answers to 1 decimal place)

$S \sim$ ($\mu_S =$, $\sigma_S =$)

and find the probability that the total weight of a sample of 28 randomly selected adult females is less than 4862 lbs.

(Round the answer to 4 decimal places)

Question Help:  [Video](#)

Question 26

 0/1 pt  3  19

What is the smallest sample size that guarantees that the margin of error is less than 3% when constructing a 99% confidence interval for a population proportion?

(Round UP to the nearest integer)

Question Help:  [Video](#)